Set 2

MORADABAD INSTITUTE OF TECHNOLOGY

APPLIED SCIENCES & HUMANITIES DEPARTMENT

SESSIONAL TEST - I

In pursuit of excellence

Course: B.Tech.
Session 2025-26
Subject Name: MATHS-IV
Max. Marks: 20

Semester: III
Section: E, F, G, I, J
Subject Code: (BAS-303)
Time: 1 Hour

Q. No. :	1	2	3	4	5	6
CO No. :	1	1	2	1	2	3
Bloom's Level	K3	K3	K3	K3	K3	K3

SECTION-A

2*1=2 Marks

Q.1. The first four moments of a distribution about x = 4 are 1, 4, 10 and 45. State whether the distribution is leptokurtic or platykurtic.

SECTION-B

3*2=6 Marks

Q.2. Ten students got the following percentage of marks in Chemistry and Physics:

Marks in Chemistry	78	36	98	25	75	82	30	62	65	39
Marks in Physics	84	51	91	60	68	62	86	58	63	47

Calculate the rank correlation coefficient.

Q.3. Three urns contain 6 red, 4 black; 4 red, 6 black and 5red , 5 black balls respectively. One of the urns is selected at random and a ball is drawn from it. If the ball drawn is red, find the probability that it is drawn from the first urn.

SECTION-C

4*3=12 Marks

Q.4. Fit the curve $y = ae^{bx}$ to the following data by the method of Least squares:

x	1	5	7	9	12
y	10	15	12	15	21

Q.5. Out of 800 families with 4 children each, how many families would be expected to have (i) 2 boys and 2 girls (ii) at least one boy (iii) no girl ? Assume equal probabilities for boys and girls.

OR

The average monthly sales of 5000 firms are normally distributed. Its mean and standard deviation are ₹ 36000 and ₹ 10000 respectively. Find the number of firms having sales between ₹ 30000 and ₹ 40000.

[Given area under normal curve from 0 to z for Z (0.4) = 0.1554 and Z (0.6) = 0.2257].

Q.6. A die is thrown 90 times with the following results:

Face	1	2	3	4	5	6	Total
Frequency	10	12	16	14	18	20	90

Use Chi-square test to check whether these data are consistent with the hypothesis that die is unbiased.

Given $\chi^2_{0.05} = 11.07$ (for 5 degree of freedom).

OR

A sample of 18 items has a mean 24 units and standard deviation 3 units. Test the hypothesis that it is a random sample from a normal population with mean 27 units.

Given $t_{0.05} = 2.11$ (for 17 degree of freedom).

Set A

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APPLIED SCIENCES & HUMANITIES DEPARTMENT

SESSIONAL TEST - I

In pursuit of excellence

Course: B.Tech.
Session 2025-26
Subject Name: MATHS-IV
Max. Marks: 20

Semester: IV
Section: A, B, C, D
Subject Code: (BAS-403)
Time: 1 Hour

Q. No. :	1	2	3	4	5	6
CO No. :	4	3	3	3	4	4
Bloom's Level	K2	K3	K3	K3	K3	K3

SECTION-A

2*1=2 Marks

Q.1. If X is a Poisson variate such that $P(X = 2) = 9P(X = 4) + 90P(X = 6)$, find the standard deviation.

SECTION-B

3*2=6 Marks

Q.2. Ten students got the following percentage of marks in Chemistry and Physics:

Marks in Chemistry	78	36	98	25	78	82	30	62	65	39
Marks in Physics	84	51	91	60	68	62	86	58	63	47

Calculate the rank correlation coefficient.

Q.3. Find the skewness and kurtosis on the basis of moments for the following data:

Class	10-20	20-30	30-40	40-50
frequency	18	20	30	22

SECTION-C

4*3=12 Marks

Q.4. Fit the curve $y = ae^{bx}$ to the following data by the method of Least squares:

x	1	5	7	9	12
y	10	15	12	15	21

Q.5. Out of 800 families with 5 children each, how many families would be expected to have (i) 3 boys and 2 girls (ii) 2 boys and 3 girls (iii) no girl. Assume equal probabilities for boys and girls.

Q.6. The lifetime of an electric bulb is normally distributed with mean 1200 hours and standard deviation 100 hours.
(i). Find the probability that a bulb lasts (a) more than 1300 hours, (b) between 1100 and 1250 hours.
(ii). Out of 2000 bulbs, how many are expected to last more than 1400 hours?

Given $P(0<z<1)=0.3413$, $P(0<z<0.5)=0.1915$, $P(0<z<2)=0.4772$

Set A

MORADABAD INSTITUTE OF TECHNOLOGY

APPLIED SCIENCES & HUMANITIES DEPARTMENT

SESSIONAL TEST - II

Course: B.Tech.
Session 2025-26
Subject Name: MATHS-IV
Max. Marks: 20

Semester: IV
Section: A, B, C, D
Subject Code: (BAS-403)
Time: 1 Hour

Q. No. :	1	2	3	4	5	6
CO No. :	2	1	2	1	3	2
Bloom's Level	K2	K3	K3	K3	K3	K3

SECTION-A

2*1=2 Marks

Q.1. Classify the partial differential equation $3\frac{\partial^2 u}{\partial x^2} - 2\frac{\partial^2 u}{\partial x \partial t} + 4\frac{\partial^2 u}{\partial t^2} - \frac{\partial u}{\partial t} = 0.$

SECTION-B

3*2=6 Marks

Q.2. Solve by **Lagrange's method** $(mz - ny)p + (nx - lz)q = ly - mx$.

Q.3. Solve by **the method of separation of variables** $4\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = 3u; \quad u(x,0) = 3e^{-x} - e^{-5x}$

SECTION-C

4*3=12 Marks

Q.4. Solve $\frac{\partial^2 z}{\partial x^2} - 3\frac{\partial^2 z}{\partial x \partial y} + 2\frac{\partial^2 z}{\partial y^2} = \sin(x - 2y)$

Q.5. The following data shows the value of sample mean $\bar{X}$ and range R for 10 samples of size 5 each. Draw $\bar{X}$ and R chart and determine whether the process is under control.

Sample no.	1	2	3	4	5	6	7	8	9	10
Mean $\bar{X}$	11.2	11.8	10.8	11.6	11	9.6	10.4	9.6	10.6	10
Range R	7	4	8	5	7	4	8	4	7	9

Assume for n=5, A_2= 0.577, D_3 = 0 and D_4 = 2.115.

OR

The following data is collected on two characters:

	smokers	Non smokers
Literate	83	57
Illiterate	45	68

Based on this information can you say that there is no relation between habit of smoking and literacy. (Given tabulated value of χ^2 at 5% level of significance for 1 d.f. is 3.841)

Q.6. A string is stretched and fastened to two points l apart. Motion is started by displacing the string in the form $y = A\sin\frac{\pi x}{l}$ from which it is released at time t=0. Show that the displacement of any point at a distance x from one end at time t is given by $Y(x,t) = A\sin\frac{\pi x}{l}\cos\frac{\pi ct}{l}$.

OR

Use Fourier sine transform to solve the equation $\frac{\partial u}{\partial t} = 2\frac{\partial^2 u}{\partial x^2}$ under the conditions

i. $u(0,t) = 0$
ii. $u(x,0) = e^{-x}$
iii. $u(x,t)$ is bounded.